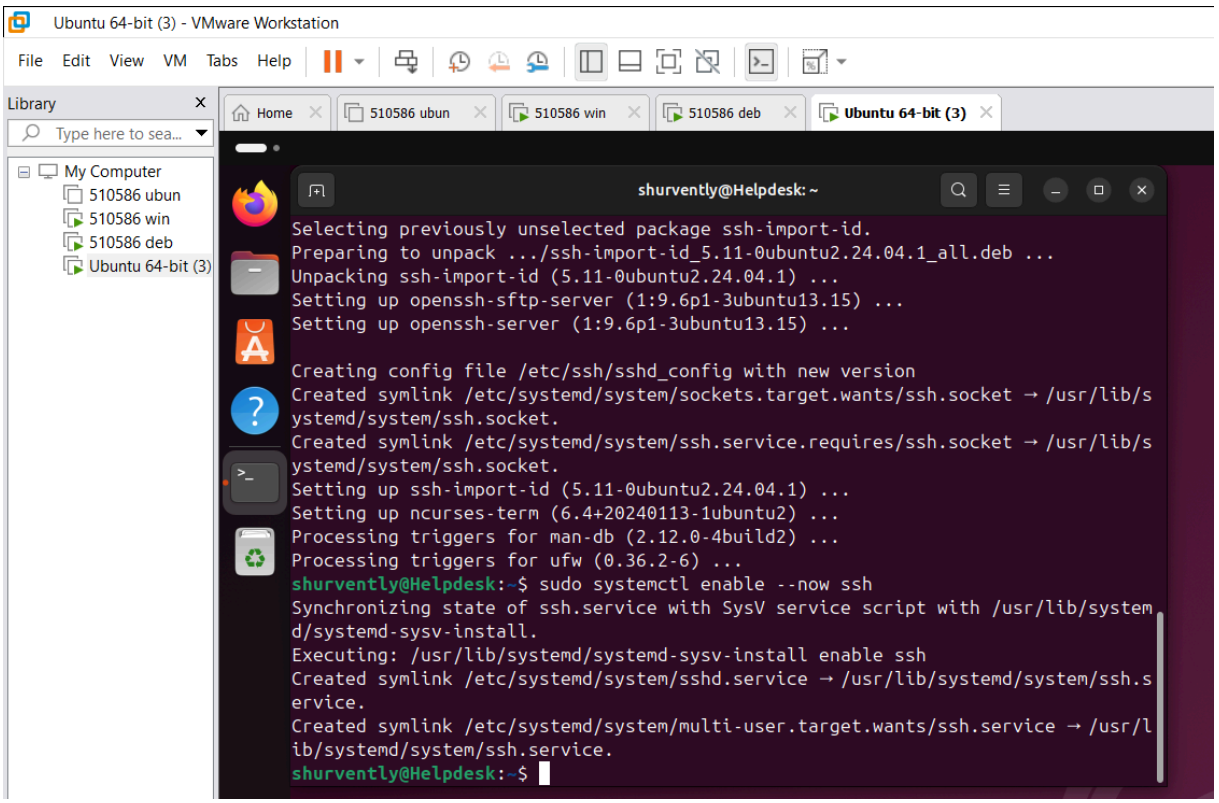
Template Week 6 – Networking

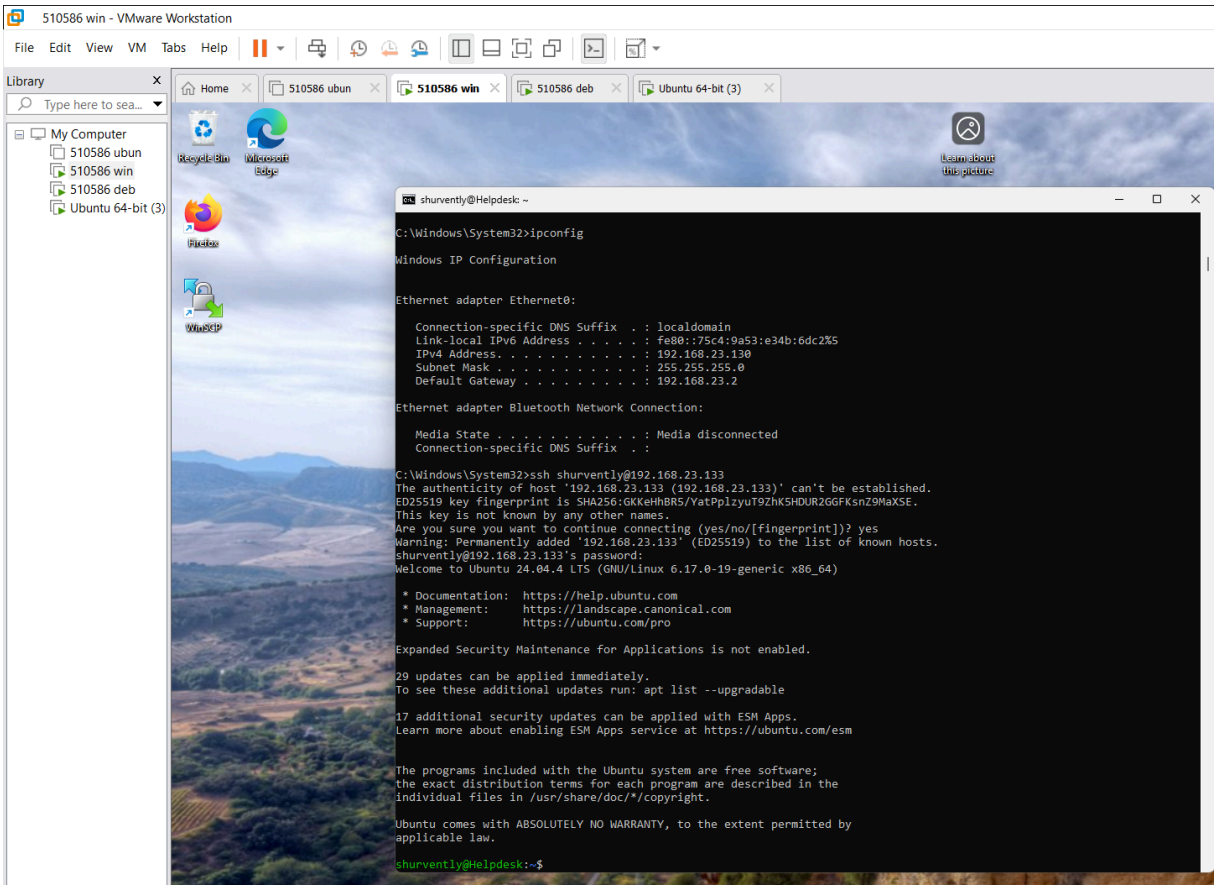
Student number:510586

Assignment 6.1: Working from home

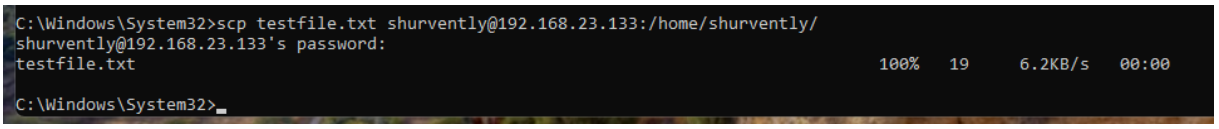
Screenshot installation openssh-server:



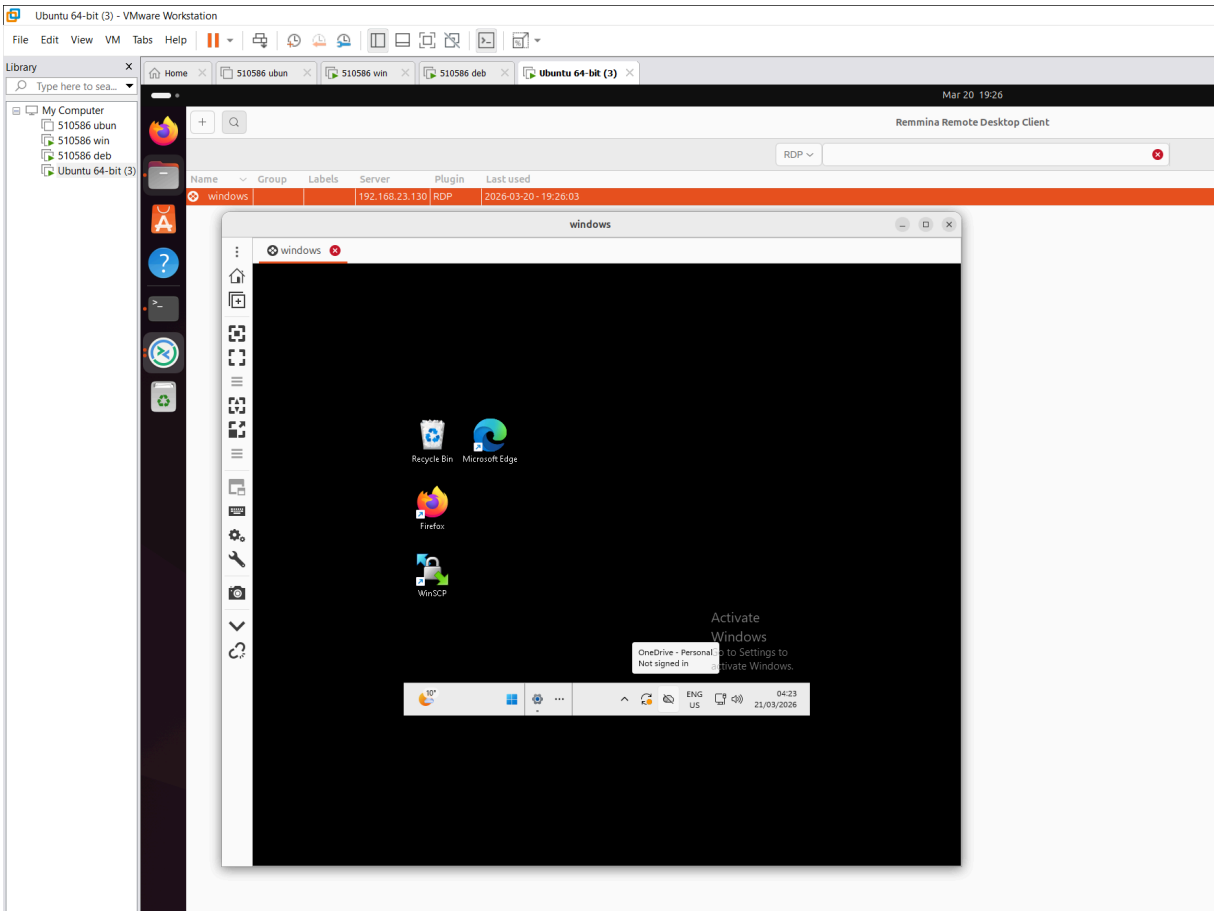
Screenshot successful SSH command execution:



Screenshot successful execution SCP command:

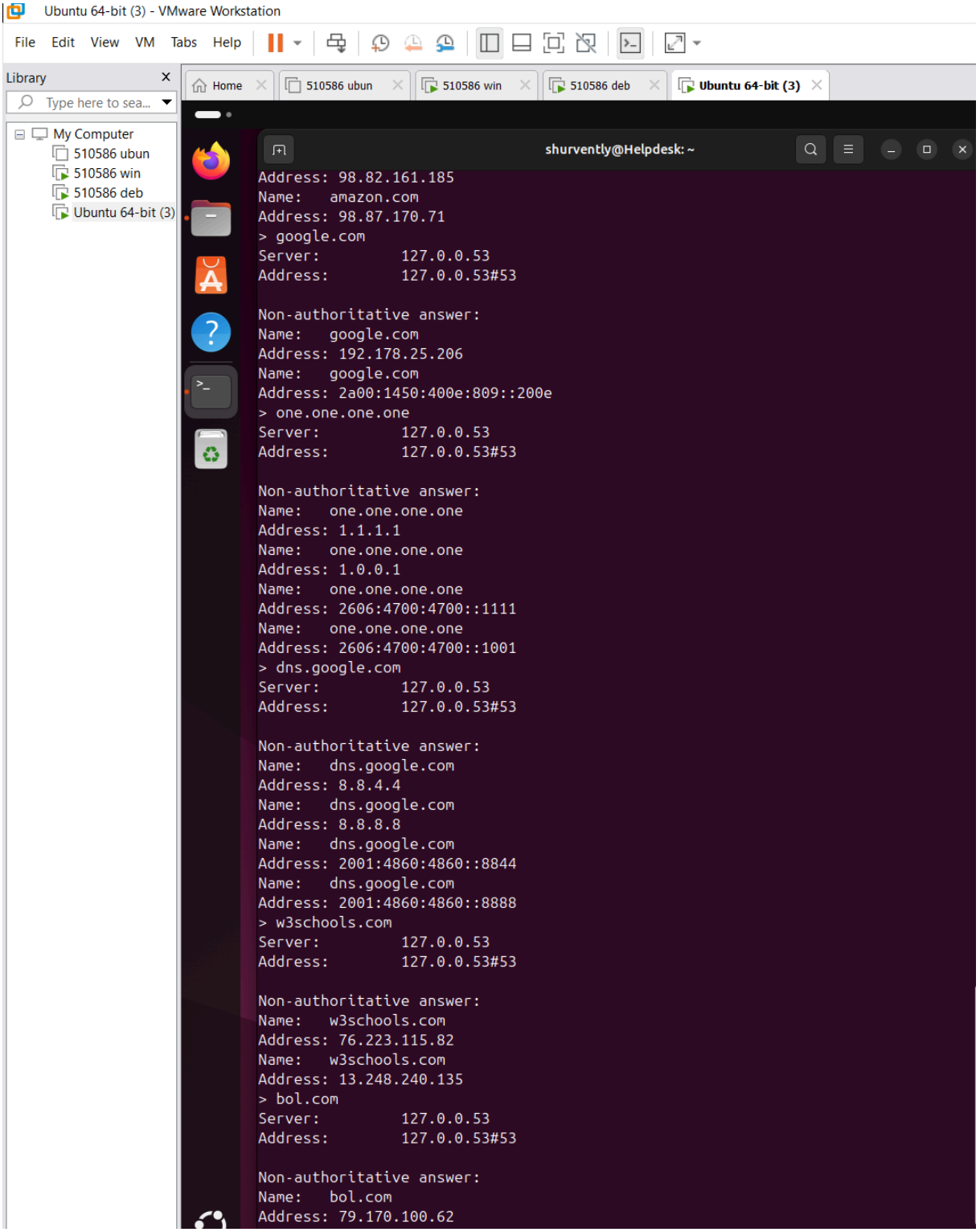


Screenshot remmina:

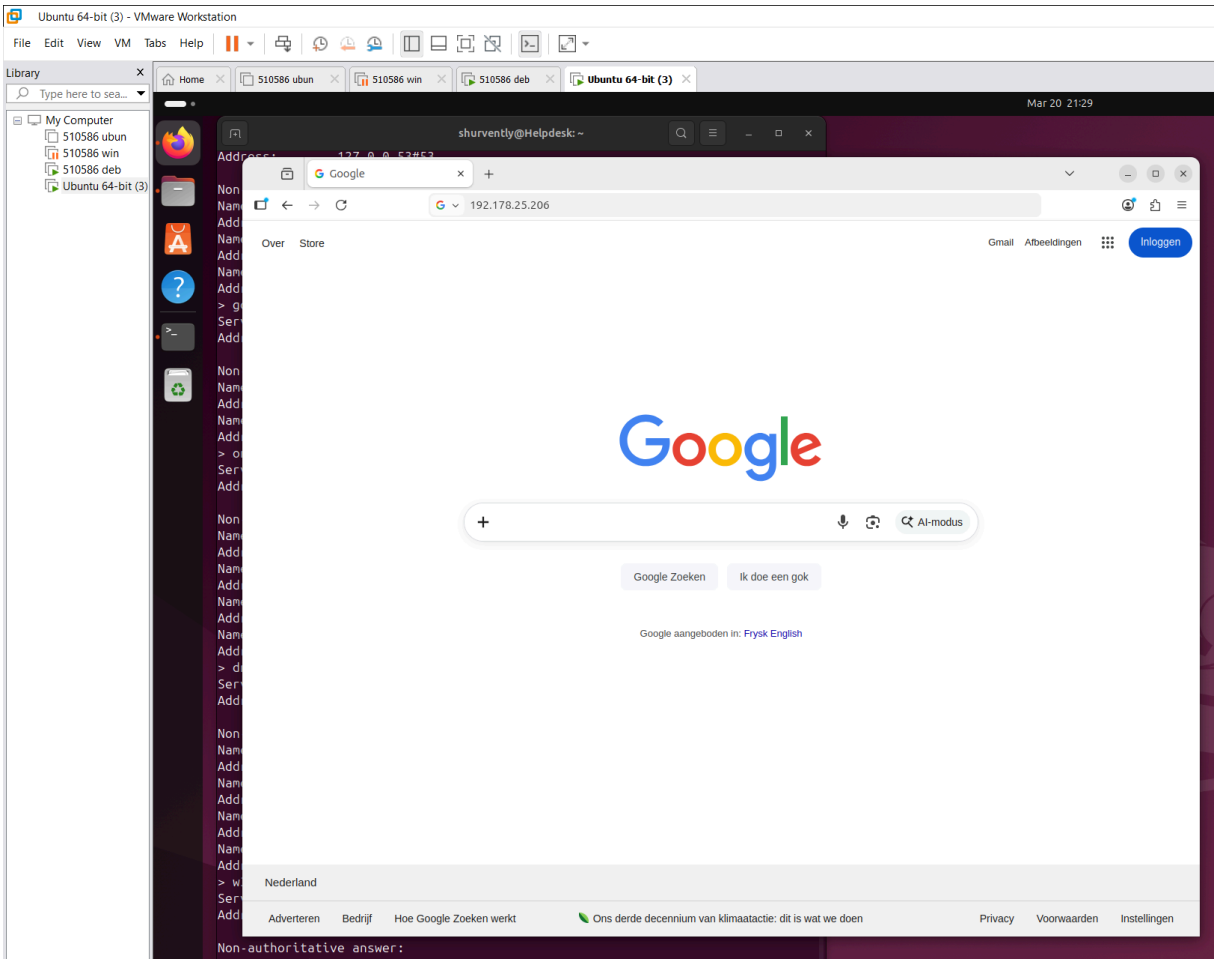


Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:


Ubuntu 64-bit (3) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
510586 ubun
510586 win
510586 deb
Ubuntu 64-bit (3)
Home
510586 ubun
510586 win
510586 deb
Ubuntu 64-bit (3)
shurvently@Helpdesk: ~
Address: 98.82.161.185
Name: amazon.com
Address: 98.87.170.71
> google.com
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: google.com
Address: 192.178.25.206
Name: google.com
Address: 2a00:1450:400e:809::200e
> one.one.one.one
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: one.one.one.one
Address: 1.1.1.1
Name: one.one.one.one
Address: 1.0.0.1
Name: one.one.one.one
Address: 2606:4700:4700::1111
Name: one.one.one.one
Address: 2606:4700:4700::1001
> dns.google.com
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: dns.google.com
Address: 8.8.4.4
Name: dns.google.com
Address: 8.8.8.8
Name: dns.google.com
Address: 2001:4860:4860::8844
Name: dns.google.com
Address: 2001:4860:4860::8888
> w3schools.com
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: w3schools.com
Address: 76.223.115.82
Name: w3schools.com
Address: 13.248.240.135
> bol.com
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: bol.com
Address: 79.170.100.62

Screenshot website visit via IP address:



Assignment 6.3: subnetting

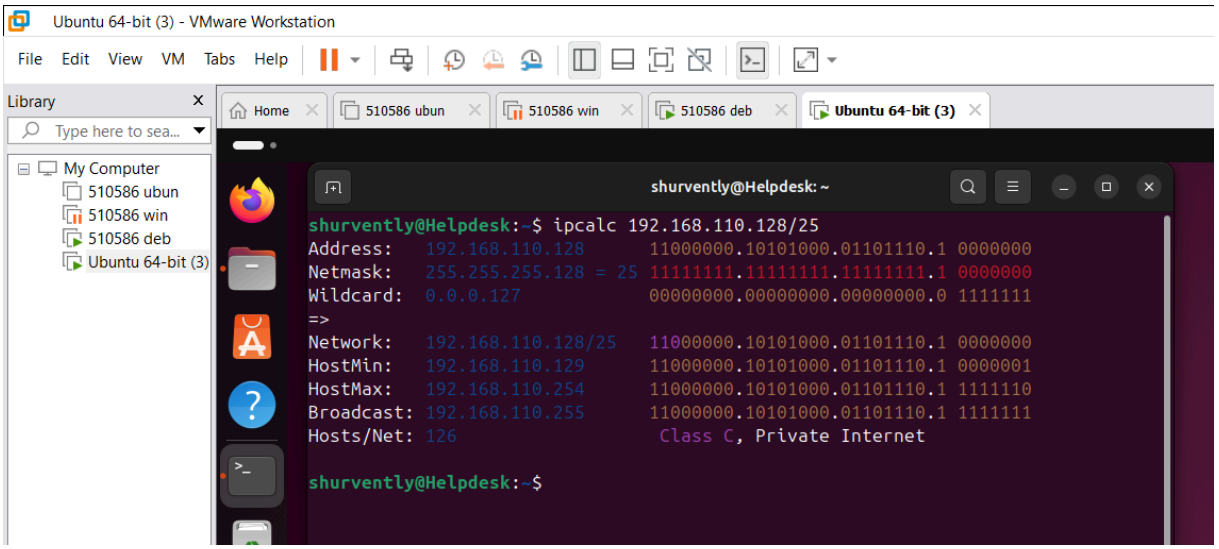
How many IP addresses are in this network configuration 192.168.110.128/25?

With a subnet of /25 there are 7 bits left for the host. Which means 2 to the power of 7 =128 ip's

What is the usable IP range to hand out to the connected computers?

1 is needed for networks and one is needed for broadcasting which means 128-2=126 ip's left to be distributed to connected computers.

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`



Explain the above calculation in your own words.

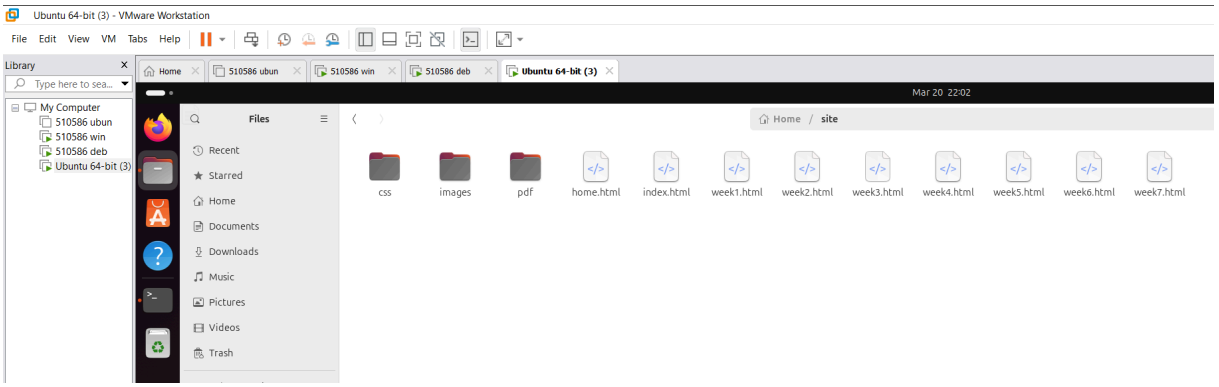
1 is needed for networks and one is needed for broadcasting which means 128-2=126 ip's left to be distributed to connected computers.

Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:

```
shurvently@Helpdesk:~/site$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4f:fd:7a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.23.133/24 brd 192.168.23.255 scope global dynamic noprefixroute ens33
       valid_lft 1382sec preferred_lft 1382sec
    inet6 fe80::20c:29ff:fe4f:fd7a/64 scope link
       valid_lft forever preferred_lft forever
```

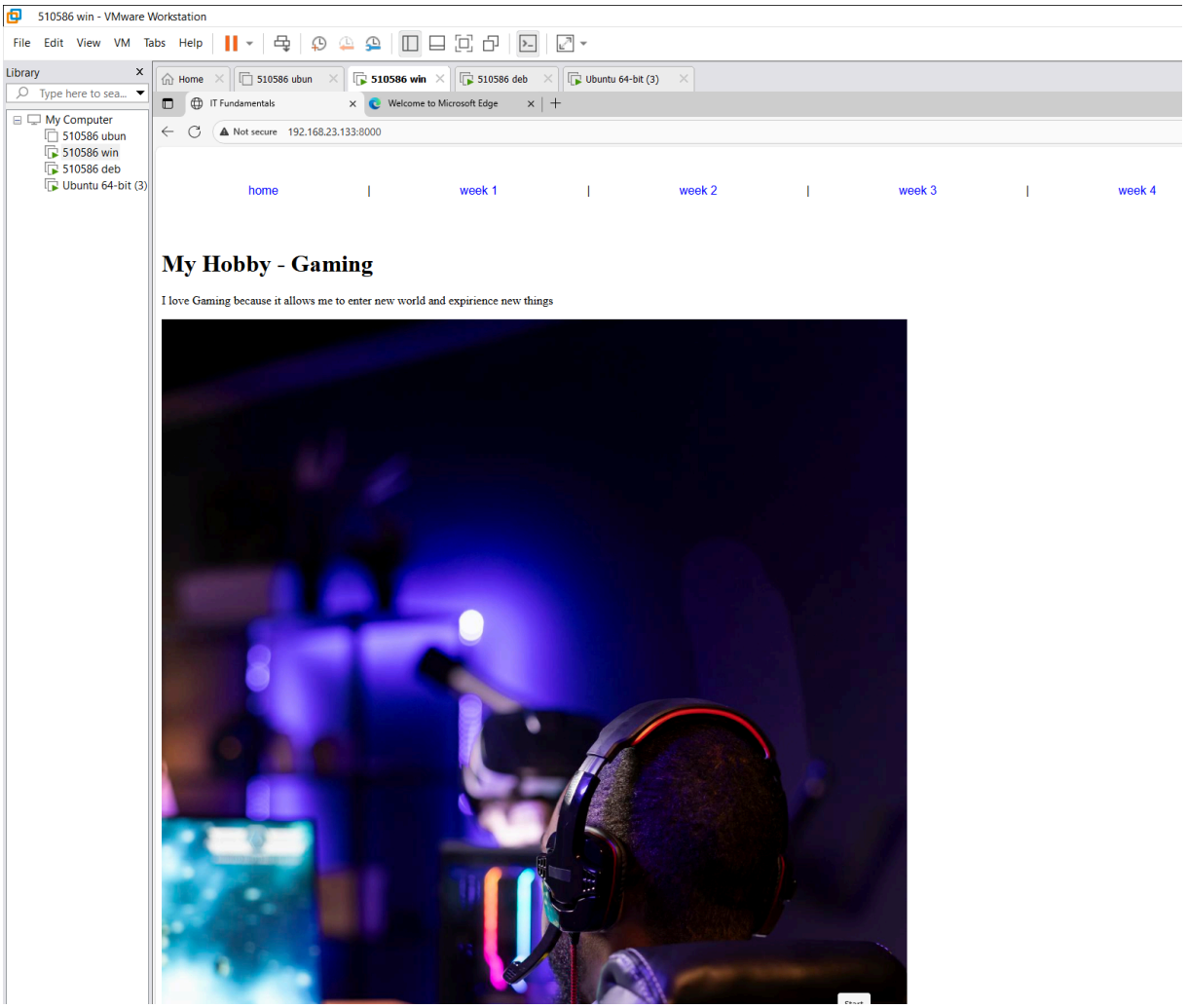
Screenshot of Site directory contents:



Screenshot python3 webserver command:

```
shurvently@Helpdesk:~/site$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.23.130 - - [20/Mar/2026 21:59:23] "GET / HTTP/1.1" 200 -
192.168.23.130 - - [20/Mar/2026 21:59:23] "GET /css/mypdfstyle.css HTTP/1.1" 200 -
192.168.23.130 - - [20/Mar/2026 21:59:23] "GET /home.html HTTP/1.1" 200 -
192.168.23.130 - - [20/Mar/2026 21:59:23] "GET /images/pic2.jpg HTTP/1.1" 200 -
192.168.23.130 - - [20/Mar/2026 21:59:23] "GET /images/pic1.jpg HTTP/1.1" 200 -
192.168.23.130 - - [20/Mar/2026 21:59:23] code 404, message File not found
192.168.23.130 - - [20/Mar/2026 21:59:23] "GET /favicon.ico HTTP/1.1" 404 -
```

Screenshot web browser visits your site



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

```
Example: 192.168.1.100/27
Calculate the network segment
IP Address:   11000000.10101000.00000001.01100100
Subnet Mask:  11111111.11111111.11111111.11100000
-------------------------------------------------
Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.
```

```
For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).
The range of this network segment is from 192.168.1.96 to 192.168.1.127.
```

Paste source code here, with a screenshot of a working application.

Ready? Save this file and export it as a pdf file with the name: **week6.pdf**